

$$\frac{dA}{dt} - V'(F) = 0$$

Euclidean condensate equation

Level 0: fundamental

$O(4) \rightarrow O(3)$ SSB

$$K_{AB} \frac{dA}{dt} \frac{dB}{dt} x + m^2 x = 0$$

$$K_{AB} = b \frac{dA}{dt} \frac{dB}{dt} - a \frac{dA}{dt} \frac{dB}{dt}$$

Ordered-branch field equation

Level 1: broken-phase EFT

Real parametrisation $t = w/c^*$

$$\frac{d^2 \phi}{dt^2} - c^2 \frac{d^2 \phi}{dx^2} + M^2 \phi = 0$$

Klein-Gordon equation

Level 2: Lorentzian branch

Nonrelativistic limit

$$i \hbar \frac{d\psi}{dt} = -\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + V \psi$$

Schrodinger equation ($\hbar \rightarrow \hbar$)

Level 3: NR coherent branch